

FINANCIAL STIMULUS AND MICRO FINANCE INSTITUTIONS*

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August 17, 2026

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Abstract

We study how microfinance institutions (MFIs) shape the allocation and effectiveness of loan guarantees during recessions. We develop a stylized model where MFIs specialize in small borrowers who are more responsive to receiving a guaranteed loan. MFIs have better information about them, yet better information does not necessarily improve welfare, because private lending incentives differ from the social planner. Thus, the optimal MFI participation in loan guarantee programs depends on their specialization and on the distorted private incentives that are amplified by better information. We then exploit a large-scale loan guarantee program for small firms implemented in Peru during the COVID-19 recession, combined with administrative data covering the universe of small business loans. Guarantees significantly reduced delinquency, particularly among micro-firms and firms in high-contact industries, segments that faced more severe liquidity needs. MFIs channeled a larger share of guarantees to these high-sensitivity segments, while traditional banks prioritized inframarginal firms. We calibrate a quantitative model with lender specialization and heterogeneous firms and show that the observed MFI participation raised the debt saved from default by 29 percent relative to a banks-only counterfactual. Further increasing MFI participation to its optimal level would raise the debt saved from default by an additional 33 percent.

JEL: G18, G21, H81

*We thank Greg Buchak, Pierre Dubuis, Dan Greenwald, Apoorv Gupta, Sean Higgins, Justin Holz, Adrien Matray, Nicola Pavanini, Jorge Ponce, Claudio Raddatz, Giorgio Sertsios, Russell Toth, Eric Zwick, and seminar participants at Central Reserve Bank of Peru, Elsevier Finance Conference, CEMLA/Dallas Fed Financial Stability Workshop, Central Bank of Brazil, Finance UC, ICBFS, Central Bank of Chile, NEUDC, FIRS, and WEAI for helpful discussions and comments. Estefano Benvenuto, Bryan Camiloaga and Edgard Oporto provided outstanding research assistance. Carlos Burga gratefully acknowledges financial support from Fondecyt Iniciación, project number 11250729. The views expressed herein are those of the authors and do not necessarily reflect those of the Central Reserve Bank of Peru.

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1 Introduction

Loan guarantees are a widely used policy tool for sustaining small business credit during recessions. A key feature of these programs is their distribution through private lenders, a design intended to prevent political interference. Yet, it remains unclear whether private lenders actually direct guarantees to the most sensitive firms, those whose survival responds most to liquidity support. Instead, they may favor inframarginal clients that would survive anyway, such as existing large borrowers, or they may shift toward riskier firms when government guarantees reduce their skin in the game.

Microfinance institutions (MFIs) play a major role in small business finance in emerging markets. Unlike traditional banks, which rely on hard information, MFIs specialize in relationship lending. This technology allows them to serve smaller, opaque firms that traditional banks often overlook, a segment that tends to be most sensitive to liquidity support.

However, better information is not a guarantee of a better allocation. When coverage is high, lenders bear little of the default risk and put little weight on borrowers' survival. Thus, MFIs' superior information can target risky borrowers at the expense of solid survivors. Moreover, MFIs distribute guarantees at higher costs, and their advantages operate on a small-size clientele. Therefore, when designing these programs, policymakers must weigh the targeting gains from MFIs' specialization and information against the distorted use of that information under high coverage, their higher origination costs, and the limited size of their clientele.

We study how lender incentives shape the allocation and impact of small business loan guarantees, and how much of these programs should be channeled through specialized lenders such as MFIs in developing countries.

We address these questions in the context of Reactiva Perú, a large-scale loan guarantee program implemented by the Peruvian government during the COVID-19 pandemic. This

setting offers two key advantages. First, the program, equivalent to 8% of GDP, operated in a financial system where the microfinance sector is mature, accounting for over half of all small-firm lending.¹ This allows us to study the role of MFIs at scale, which is crucial to understanding the transmission of micro-credit shocks (Breza and Kinnan 2021). Second, comprehensive regulatory reporting provides loan-level data on firm debt across traditional banks and MFIs, enabling us to measure credit balances and guarantee allocation by lender type. Together, these features allow us to identify the causal effects of loan guarantees and quantify how MFI participation shapes their allocation and effectiveness.

We use administrative data covering the universe of small-firm lending relationships in Peru from 2019 to 2021. For each lender-firm pair, we observe outstanding loan balances, delinquency (our empirical proxy for firm failure), and city of loan origination. Firm-level characteristics include industry, age, and lender-reported risk rating. We merge these financial data with annual firm tax records containing sales, wage bill, and capital stock for all formally registered small firms. The granularity of our data allows us to trace which firms received guarantees and from which type of lender, directly linking lender composition to program effectiveness.

Our analysis proceeds in three steps. First, we develop a conceptual framework to illustrate the channels through which the mix of lenders matters for the effectiveness of loan guarantees and to identify the moments that discipline them. Second, we estimate these moments in the data and show that MFIs channeled a larger share of resources to the most sensitive firms (e.g., small firms in industries with high in-person contact with customers, such as restaurants or retail), while traditional banks served inframarginal borrowers that would have survived without assistance (e.g., large firms), consistent with their ex-ante specialization. Third, we feed these estimates into a calibrated quantitative model with lender

¹Peru represents a leading example of microfinance development, ranking consistently among the top markets for MFIs according to the Inter-American Development Bank. See, for example, https://graphics.eiu.com/assets/images/public/Microscope_on_Microfinance_2014/EIU-Microscope-Dec-2015.pdf.

specialization and heterogeneous firms to quantify the optimal participation of MFIs.

In the first part of the paper, we build a conceptual framework to illustrate how MFIs shape the allocation of loan guarantees. We consider a continuum of heterogeneous firms, each characterized by revenue and debt obligations that jointly determine net cash-flow. We model the recession as a firm-specific, unexpected liquidity shock. Firms survive if their net cash-flow, plus any guaranteed loan received, exceeds the shock. Firms differ in opacity, with opaque firms being more likely to have low net cash-flow, so they are more sensitive to liquidity support than transparent borrowers.² Two types of lenders, MFIs and traditional banks, serve these firms and differ along three dimensions. First, MFIs' portfolios are tilted toward opaque borrowers, which defines a *specialization channel*. Second, their relationship-based screening technology allows them to observe the revenue of opaque firms with a higher probability than banks, which introduces a *precision channel*. Finally, MFIs distribute guarantees at a higher origination cost, which we call the *cost channel*.

We define the social value of a guarantee as the debt it saves, net of default losses on the guaranteed loan, origination costs, and the fiscal cost of the guaranteed-loan default payout. The social planner therefore ranks firms by a combination of the treatment effect, defined as the increase in survival probability generated by the guaranteed loan, and the survival probability of treated firms. The social planner allocates guarantees to the highest-value firms until the budget is exhausted.

Treatment effect and survival probability of treated firms are different objects. A fragile, low-cash firm may exhibit a large treatment effect and still be likely to fail. On the other hand, a solid firm that would probably survive regardless of a guarantee has high treated survival probability but a small treatment effect. The planner values both because a large treatment effect alone saves little debt when the probability of survival remains low.

²This assumption is consistent with the treatment-effect heterogeneity we estimate in the data. In our quantitative analysis, however, we endogenize the firm-size distribution and lender specialization, so the treatment-effect heterogeneity across lenders emerges as an equilibrium object.

Private lenders allocate guarantees weighting the same two objects as the planner, the treatment effect and treated survival, but with different weights. Furthermore, private lenders can individually rank only clients they observe, but pool the firms they do not observe, treating unobserved opaque borrowers as an average client. MFIs' technology that allows them to observe the revenue of opaque firms with higher probability therefore translates into a finer ranking of opaque borrowers relative to banks.

Furthermore, the loan guarantee coverage rate moves the weights of private lenders and the social planner in opposite directions. A higher coverage insulates banks and MFIs from default losses, so they weight survival much less, and become willing to provide loan guarantees to high-treatment but fragile firms over solid survivors. On the other hand, as coverage increases, every default becomes more expensive for the government, so the planner's weight on treated survival rises. We show that there is a unique coverage threshold at which both rankings coincide.

Above the coverage threshold, MFIs' screening technology that allows them to act on borrower-level comparisons that traditional banks cannot, may promote high-treatment but fragile clients at the expense of solid survivors. Thus, better information does not necessarily improve welfare. Overall, the welfare contribution of a marginal guarantee channeled through MFIs is the sum of three forces: specialization, precision, and a cost gap; and the welfare-maximizing MFI participation is a quantitative question.

In the second part of the paper, we identify two moments that map into the channels of our conceptual framework: which firms are the most sensitive to guarantees, and whether they are primarily targeted by MFIs or banks. To do so, we estimate the causal effects of the program using a difference-in-differences design that exploits cross-sectional variation in lenders' guarantee uptake. We compare credit supply from high- and low-treatment lenders to the same firm before and after the program. The key identifying assumption is that, absent the program, credit supply would have evolved in parallel trends across lenders with

different treatment intensities.

We provide evidence supporting our identifying assumption in three ways. First, we plot event-study graphs showing no evidence of pre-trends. Second, we control for time-varying, lender-size-dependent shocks by comparing financial institutions within the same size quartile, which mitigates the concern that endogenous bank-firm matching drives the results. Third, we exploit a three-month delay in the program’s implementation as a placebo test, documenting that our treatment measure is uncorrelated with credit during the initial COVID-19 restrictions, before guarantees were distributed. We complement this evidence with a covariate balance test showing that treatment is uncorrelated with predetermined lender-level delinquency rates, leverage, ROA, and foreign-liability dependence.

We find that financial institutions with one standard deviation higher treatment increased credit supply by 10 percent while reducing non-guaranteed lending by 30 percent, suggesting partial crowding out.

We use a difference-in-differences instrumental variable approach to estimate the firm-level effects of the program. We define a firm-level instrument equal to the weighted average bank-level treatment, with weights given by the pre-existing shares of each lender in firm total debt. We find that, for the average firm, a 10 percent credit expansion reduces delinquency by 1.4 percentage points, and receiving a guaranteed loan lowers delinquency by 11 percentage points. These average effects mask substantial heterogeneity across lenders’ borrower pools. The average MFI client participating in the program experiences a 15 percentage point decline in delinquency probability, while the average traditional bank client shows no statistically significant change. We document that firms participating in the program experienced an 80 percent credit expansion, similar across borrowers served by traditional banks and MFIs.

To identify the most sensitive borrowers and study the lender-specific allocation of loan guarantees, we use two proxies for liquidity needs during the COVID-19 recession, namely

firm size and industry exposure to in-person contact. A 10 percent credit increase reduces delinquency by 1.5 percentage points for the smallest firms (bottom 90% of the debt distribution), compared to only 0.4 percentage points for larger small firms (top 10%).³ Similarly, a 10 percent credit expansion reduces delinquency by 1.6 percentage points for firms in high-contact industries, versus 0.9 percentage points for the remaining firms. Our results indicate that guarantees helped firms alleviate the liquidity needs generated by the pandemic restrictions.

Having identified the most sensitive firms, we then compare the lending patterns of MFIs and traditional banks. We document that MFIs prioritized sensitive borrowers, allocating 65 percent of their guarantees to the smallest firms, while traditional banks allocated only 29 percent. Furthermore, 60 percent of MFIs' guarantees went to firms in industries with high exposure to in-person contact with customers (e.g., retail, restaurants, transportation), compared to 46 percent for traditional banks. A back-of-the-envelope calculation suggests that, under the actual mix of lenders, the program reduced delinquency by 12 percentage points, compared to 8 percentage points if only traditional banks distributed the guarantees.

These allocation patterns closely mirror lenders' pre-recession specialization. Prior to the pandemic, traditional banks concentrated 74 percent of their small-firm lending in the largest borrowers, while MFIs allocated only 50 percent of their portfolios to this segment. The same pattern holds by industry: MFIs held 64 percent of their pre-recession portfolios in high-contact industries, versus 59 percent for traditional banks.

We then explore how the effects on sensitive borrowers vary by the type of lender that provided the guarantee. We find that, among highly sensitive firms, those that received guarantees from MFIs performed better than those served by traditional banks. A 10 percent increase in credit reduced delinquency by 4 percentage points for smaller firms served by MFIs, compared to only 0.3 percentage points for those served by banks. Similarly, a 10 per-

³We consider companies within the "small firm" category. All comparisons are within this segment.

cent credit increase reduced delinquency by 4 percentage points for high-contact firms served by MFIs, while the response for similar firms served by traditional banks was statistically insignificant.

These estimates map into the channels of the conceptual framework. The allocation shares reflect that MFIs' portfolios are tilted toward sensitive borrowers (specialization), while the larger effects among sensitive firms served by MFIs are consistent with MFIs' guarantees reaching the more responsive borrowers (precision). The reduced form, however, can neither separate the two channels nor predict how allocation would change under a different mix of lenders. We address both questions using a quantitative model disciplined by these estimates.

In the third part of the paper, we develop a quantitative model featuring heterogeneous firms, who choose whether to operate in the formal economy or not, and two lender types, MFIs and traditional banks, that compete in prices under differentiated product demand and can screen informal borrowers at a cost. Lenders differ in their screening technology. MFIs can observe the revenue of opaque borrowers at a lower cost than traditional banks, and choose optimally which firms to screen, while both lenders read formal borrowers equally well. This technology, together with a higher loan origination cost of MFIs, generates distinct client distributions for each lender in equilibrium.

Firms draw a productivity signal and decide whether to pay a fixed cost to become formal, hard-information firms, or to remain informal and opaque at zero cost. Then, they commit to a loan size, fully financed with debt, before a transitory productivity shock is realized. Leverage chosen under such uncertainty becomes the source of default in normal times. Finally, firms choose among lenders subject to idiosyncratic taste shocks, which determines credit demand. As in the conceptual framework, the recession is modeled as an unexpected liquidity shock that hits firms, and the government responds injecting guarantees distributed by private lenders.

The calibrated model matches key empirical moments of the data, including the observed specialization of lenders and the reduced-form treatment effects. MFIs' clients are on average smaller and informal, while traditional banks' clients are larger and formal. Informal lenders pay higher interest rates due to opacity and screening costs, and MFIs charge higher interest rates than banks due to their specialization in informal borrowers.

The model delivers a sharp decomposition of the MFI advantage. Across all potential recipients, the average treatment effect is roughly equal between the two clienteles, around 10 percentage points. However, among firms that actually receive guarantees, the average treatment effect is 20 percentage points for MFI recipients versus 11 for bank recipients. The MFI advantage therefore arises from which clients receive loan guarantees, not from their clients being uniformly more responsive.

We evaluate the program with two main measures. First, the total debt saved from default by the program, which is the Peruvian government stated goal. Second, a welfare measure that nets the debt saved against default losses, origination costs, and the fiscal cost of defaulting guarantees. Under the observed MFI participation, the program saved 4.8 percent of aggregate outstanding debt, compared to 3.7 percent in a counterfactual where only traditional banks distributed guarantees. The MFI participation that maximizes this metric is 64 percent, far above the observed participation, reaching 6.3 percent of aggregate debt saved from default. The net effect of MFIs' precision channel is positive, albeit small, accounting for 5 percent of the total debt saved from default.

Our welfare measure weights additional forces such as the fiscal cost and capital losses on defaulting guaranteed loans, and lenders' guarantee-origination costs. These forces barely move the welfare-maximizing participation, which drops slightly to 62 percent, because MFI guarantee recipients survive at least as often as bank recipients. Overall, our paper shows that who distributes financial stimulus is a first-order design margin that determines the allocation of loan guarantees and, through this channel, the program's aggregate effectiveness.

Related Literature

This paper contributes to three strands of literature. First, it connects to research on government credit guarantees, a common tool for sustaining small business lending during crises (Lelarge et al. 2010; Brown and Earle 2017; Mullins and Toro 2018; Bachas et al. 2021; González-Uribe and Wang 2022; Bonfim et al. 2023; Barrot et al. 2024; Core and De Marco 2024). While existing work estimates the average effects of such programs, we study their optimal design. In this line, our paper relates to the theory of public interventions in credit markets under asymmetric information (Philippon and Skreta 2012; Tirole 2012) and, more recently, to work on the design of business credit programs during the pandemic recession (Hanson et al. 2020; Crouzet and Tourre 2021; Gourinchas et al. 2025; Elenev et al. 2022). These papers characterize the optimal instrument and the cost of intervening through a single class of intermediaries. We add lender heterogeneity and show how lenders' incentives and information shape the allocation of guarantees, which makes the composition of participating lenders itself a design object.

We also contribute to recent work examining how private intermediaries distribute financial stimulus (Haas-Ornelas et al. 2021; Joaquim and Netto 2022; Balyuk et al. 2021; Chodorow-Reich et al. 2022; Erel and Liebersohn 2022; Griffin et al. 2023). Relative to this work, our setting allows us to compare two distinct lender types, traditional banks and MFIs, that play a key role in small business credit in emerging markets, and to link their allocation to borrower sensitivity. This connects our work to the public finance literature on whether policies reach marginal instead of inframarginal firms (Zwick and Mahon 2017; Criscuolo et al. 2019; Hendren and Sprung-Keyser 2020; Slattery and Zidar 2020; Kleven 2021; Cingano et al. 2025), and to the literature on lender specialization (Blickle et al. 2026; Paravisini et al. 2023; Buchak et al. 2018).

Second, our paper contributes to the literature on microfinance institutions in emerging

markets. While early randomized controlled trials found modest real effects of microcredit (Angelucci et al. 2015; Augsburg et al. 2015; Banerjee et al. 2015b; Tarozzi et al. 2015; Attanasio et al. 2015), recent work highlights that general equilibrium forces (Kaboski and Townsend 2011; Buera et al. 2020) and large-scale interventions (Bruhn and Love 2014; Breza and Kinnan 2021; Bari et al. 2024) can generate significant aggregate impacts.

Our contribution to this literature is threefold. First, we identify a novel channel through which MFIs affect the economy: by shaping the allocation of financial stimulus during a recession. We show that MFIs are effective at distributing government guarantees to highly sensitive, smaller borrowers that traditional banks overlook, thereby strengthening the program’s aggregate impact. This finding relates to the literature on distorted private lender incentives (La Porta et al. 2003; Claessens et al. 2008; Agarwal et al. 2016). Second, we combine detailed administrative data on the universe of MFI operations with a quasi-experimental design to trace the causal effects of a large-scale financial intervention. Third, we integrate these findings into a framework where the coverage rate of the program determines how lender incentives and information shape the allocation of guarantees. We use this framework to measure the optimal participation of MFIs in loan guarantee programs.

Third, our paper contributes to the literature on information in credit markets. Seminal work studies how bank information shapes competition, relationships, and the segments each lender serves (Diamond 1984; Sharpe 1990; Rajan 1992; Petersen and Rajan 1995; Holmstrom and Tirole 1997; Hauswald and Marquez 2003, 2006; Dell’Ariccia and Marquez 2004). We study how a guarantee program changes the value of that information. Generous coverage tilts the private ranking away from the social one, so the same screening technology can improve or reduce welfare depending on the ranking that uses it. To our knowledge, this interaction between guarantee design and lenders’ information is new to this literature.

The remainder of this paper is organized as follows. Section 2 describes our data and the institutional background, and section 3 presents our conceptual framework. Section 4

discusses our empirical approach and we report our main results in section 5. Section 6 presents our model and section 7 concludes.

2 Data and Institutional Background

2.1 Data

We combine two administrative datasets covering financial and real outcomes for the universe of small firms in Peru.

1. Credit registry data. This is loan-level data from the *Reporte Crediticio de Deudores*, provided by the Central Reserve Bank of Peru. This quarterly panel spans 2019–2021 and includes outstanding loan balances, days past due, and the city of loan origination for each firm-bank relationship. Firm-level characteristics include industry classification, lender-reported credit risk ratings, and the year of first loan origination. On the lender side, we observe a unique ID used for bank regulation purposes, the type of lender (traditional bank or microfinance institution) and balance sheet information. We use this data to estimate the effects of government guarantees on credit and delinquency rates.

2. Tax reports data. This dataset contains firm-level tax records from 2018 to 2022, including annual sales, capital stock, and employment. It also includes a unique firm tax ID, enabling precise merging with credit registry data. Geographic location and industry classification are also observed, which we use to control for local economic conditions and sector-specific trends when estimating the policy’s effects.

2.2 The Peruvian Credit Market

The Peruvian banking sector has 52 financial institutions offering business loans: 15 traditional banks and 37 microfinance institutions⁴ (MFIs). Loans are categorized into five segments based on firm sales and outstanding debt: micro-credit, small-firm credit, medium-size firm loans, large firm loans, and corporate loans. For example, micro-credit is defined as loans to firms whose total debt in the banking sector is below \$6 thousand. On the other hand, loans to corporations are defined as credit to firms whose total sales in the past two years were above \$60 million.

Table 1 provides summary statistics of lender characteristics as of December 2019. The size distribution is highly skewed: the average credit portfolio is \$1 billion, while the median is \$169 million (columns 1–2). The market is also highly concentrated, with the top 5 lenders holding 80 percent of total credit (columns 3–4). MFIs account for 13 percent of aggregate lending but play a predominant role in small business loans where they originate 68 percent of micro-credit and 47 percent of small firm credit. Crucially, we focus on micro and small firm credit, which are the segments where MFIs compete with traditional banks, and call them small business loans throughout the paper.

Lender characteristics vary substantially across credit segments, as reported in the bottom panel of Table 1. While the average lender provides \$77 million of micro-credit (around three times the median of \$28 million), the average value of corporate loans is \$1.3 billion, more than seven times the median of \$166 million. Both patterns reflect the greater right-skewness of larger loan segments.

⁴Micro-finance institutions are regulated by the Peruvian Bank Supervisor but are not classified as banks. They include saving and loan institutions, financial enterprises, and enterprises for the development of small and micro firms.

Table 1: Lender Characteristics by Loan Type

	Total Loans Mean (1)	Median (2)	Number of Banks (3)	Share Top 5 Banks (4)	Share of MFIs (5)
Total	1 106	169	52	77	12.9
<i>Loans to:</i>					
Micro-credit	77	28	42	58	68.2
Small firms	190	50	45	56	47.3
Medium-size firms	263	13	48	86	5.8
Large firms	491	8	27	87	0.3
Corporations	1 272	166	13	94	0.5

This table reports bank-level summary statistics as of December 2019. We report the mean and median of the distribution of total loans across banks for each segment of business loans. Total loans are expressed in USD million. Shares are expressed as percentages.

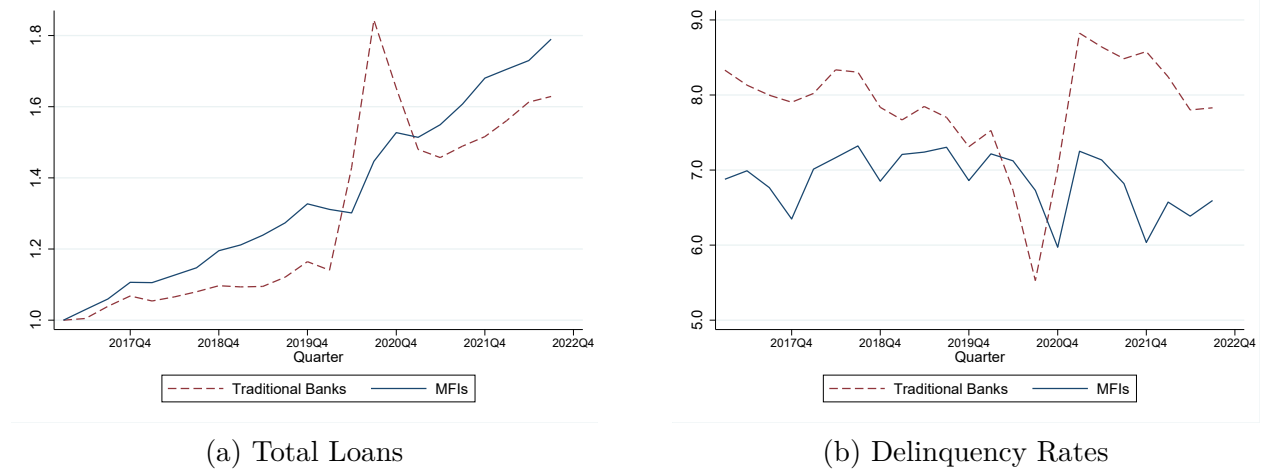
Table E1 reports summary statistics for firms with positive debt in December 2019. Column (1) and (2) reveal similar skewness in firm debt. The average value is around \$6 thousand, while the median is around \$500. Column (3) shows the share of delinquent firms (more than 30 days of repayment delay) equal to 12 percent in December 2019. Finally, column (4) shows that our sample is composed of around 3 million of firms, most of them in the segment of micro-credit (around 2.3 million).

2.3 Traditional Banks and Micro-Finance Institutions

MFIs expanded significantly in Peru over the past two decades, driven by broad deregulation. The industry is mature and plays a key role in the segment of small business loans nowadays. Figure 1 plots credit growth and delinquency rates for small business loans between 2018 and 2022. Panel (a) shows credit growth rates, measured as the value of credit in a given point in time relative to its value in 2017Q1. Before the recession, MFI credit grew 35 percent cumulatively, outpacing traditional banks' 20 percent growth. Panel (b) shows that

the rapid expansion of microfinance institutions was coupled with stable delinquency rates of 7%, slightly below traditional banks' delinquency.

Figure 1: Small Business Credit Growth and Delinquency by Type of Lender



This figure plots the evolution of small business credit and delinquency rates for traditional banks and microfinance institutions in the segment of small business loans. Panel (a) plots credit growth rate measured by the value of credit in a given point in time relative to the corresponding value in January 2018. Panel (b) plots delinquency rates measured by the share of outstanding debt with more than 30 days of repayment delay.

Table E2 compares the two types of lenders. Columns (1) and (2) consider traditional banks, while columns (3) and (4) consider MFIs. Additionally, the top panel considers the whole activities of lenders, while the bottom panel only considers small business loans. Traditional banks are larger, with 20 times bigger assets and 16 times more credit. The average traditional bank serves more cities but exhibits higher geographical concentration.⁵

MFIs are more profitable, especially the top 10 with a ROA of 1.7 percent, reported in columns (5) and (6). The average top 10 MFI attends more markets than the average traditional bank, and also exhibits a much lower geographical concentration. Considering only small business loans, the average top 10 MFI provides \$360 million in loans, while the

⁵We compute the geographical concentration as the Herfindahl-Hirschman index of bank portfolios across locations.

average traditional bank provides \$420 million.

Figure F1 maps the presence of traditional banks and MFIs across Peruvian cities as of December 2019. Around 80 percent of cities are *financial deserts*, and only 9 percent of banked locations have only traditional bank branches. In contrast, one third of banked cities are served exclusively by MFIs. While the average city served by traditional banks has 2,442 individuals per square kilometer, the average city served exclusively by MFIs has 243 individuals per square kilometer, roughly one tenth the density.⁶ Thus, MFIs play a key role in expanding geographical credit access, reaching small and opaque borrowers.

2.4 The Program of Loan Guarantees

Reactiva Perú, launched in May 2020 by the Ministry of Finance and the Central Reserve Bank, provided business loan guarantees to help firms dealing with Covid-19 restrictions. The Ministry of Finance provided the collateral, while the Central Bank provided the required liquidity. Of 52 eligible lenders, 28 participated in the program.

Guarantees covered between 80 and 98 percent of loan value (higher for smaller loans). They were allocated via first-price auctions where lenders bid on interest rates, with separate auctions for each loan type. Private lenders were in charge of screening borrowers and allocating loan guarantees. These loans had an average duration of 36 months and were granted between May and December 2020. The repayment period started 12 months after the loan was originated. Firms with poor credit rating, exhibiting more than 60 days of repayment delays in 2019, were not allowed to participate.

⁶The average financial deserts has 69 individuals per square kilometer, the average city with only bank branches has 1,407 individuals, and the average city with both lenders has 2,612 individuals per square kilometer.

Table 2: Guaranteed Loans by Type of Credit

	<u>Guaranteed Loans</u>		<u>Benefited Clients</u>		<u>Guaranteed</u>
	Value	Ratio	Number	Ratio	Rate
	(1)	(2)	(3)	(4)	(5)
Total	15.5	29	473.1	16	91
<i>Loans to:</i>					
Micro-credit	1.2	37	319.9	14	97
Small firms	3.6	42	121.8	22	95
Medium-size firms	5.9	46	28.8	81	91
Large firms	4.5	34	2.6	82	85
Corporations	0.4	3	0.2	36	80

This table reports summary statistics of guaranteed loans in different segments of the business loan market. Column (1) reports the value of loans, in USD million, distributed by the program, and column (3) reports the number of clients, in thousand, obtaining a guaranteed loan. Columns (2) and (4) are ratios computed relative to the corresponding value as of December 2019. Column (5) shows the share of the value reported in column (1) that is guaranteed by the program.

Table 2 provides summary statistics on the program's scale for different types of credit. Column (1) shows the value of guarantees distributed, and column (2) shows the ratio of this value relative to total credit in December 2019, before the recession. Similarly, column (3) shows the total number of clients attended by the program, and column (4) shows the ratio of this number relative to the total number of borrowers in 2019. The program provided \$16 billion of guaranteed loans (29 percent of 2019 credit) and reached 473 thousand firms (16 percent of borrowers). In micro-credit, the program represented 37 percent of outstanding debt but only benefited 14 percent of borrowers, while in large firm loans, the program served 82 percent of eligible firms and represented 34 percent of outstanding credit. Guarantee rates varied by loan size, from 97 percent on micro-credit to 80 percent on loans to corporations.

Regarding lender participation, traditional banks dominated the majority of auctions at the beginning, even in the segment of small business loans. This was explained by MFIs' higher operational costs, which made their bids uncompetitive. To ensure small-firm access

to loan guarantees, the Central Bank introduced MFI-specific auctions, expanding MFI participation in the program.

We can observe in Figure 1 that, while traditional banks expanded small business loans by 50 percent peak-to-trough around the program implementation, MFIs grew only 20 percent and with a lag. Credit growth converged among both types of lenders one year after the program. Furthermore, delinquency fell sharply during the one-year grace period, following the expansion of credit. However, it rose in the post-grace period, with traditional banks reaching a 3 percentage points higher delinquency relative to MFIs.

These patterns suggest that MFIs allocated guarantees more efficiently to sensitive borrowers, a hypothesis we test rigorously in the following sections.

3 Conceptual Framework

We develop a stylized model to illustrate how the composition of lenders participating in a guarantee program shapes the allocation and aggregate effectiveness of financial stimulus.

3.1 Firms

There is a continuum of firms indexed by cash-in-hand y_i and outstanding debt obligations b_i , so net cash is $n_i = y_i - b_i > 0$. Firms differ in opacity $o_i \in \{h, \ell\}$, which can be high (opaque) or low (transparent), respectively. While outstanding debt is public information, revenue is fully observable only for transparent firms. The probability that opaque firms' revenue is revealed depends on lenders' ability to gather soft information. We denote by $G_o(n | b)$ the distribution of net cash conditional on outstanding debt among firms of opacity o .

The recession is an unexpected firm-specific liquidity shock $\nu_i \geq 0$ drawn from an observable continuous distribution Φ with density ϕ . Firm i survives if $n_i + \mathbf{1}_i \varphi b_i \geq \nu_i$, where $\mathbf{1}_i$ indicates receipt of a guaranteed loan of size φb_i , with $\varphi > 0$ set by the program. Let

$s_0 = \Phi(n)$ denote the survival probability absent the program, $s_1 = \Phi(n + \varphi b)$ the survival probability with a guaranteed loan, and $T = s_1 - s_0$ the treatment effect.

Assumption 1 (Sensitivity) *For every level of outstanding debt, opaque firms are more likely to have low net cash: $G_h(n | b) = G_\ell(n + \Delta | b)$ for every b , with $\Delta \geq 0$.*

We call a recession *moderate* if the shock density ϕ is strictly decreasing on its support, so that T is strictly decreasing in n and low-cash firms respond most strongly to liquidity. Under Assumption 1, opaque firms are then more sensitive to guarantees. The treatment-effect estimates we present in the empirical section show larger effects for smaller, lower-net-cash firms, consistent with the moderate case, which we therefore adopt throughout the framework.

3.2 Lenders and the Program of Loan Guarantees

Private lenders indexed by j are risk-neutral and grouped in two types of equal clientele size: traditional banks and microfinance institutions. The government provides a total guarantee capacity Q , of which a fraction γ_M is channeled through MFIs. If a firm defaults, the government repays a fraction g of the guaranteed loan and lenders recover a fraction δ of non-guaranteed debt.

Lenders allocate guarantees among their pre-existing clients at interest rate r_g . A surviving firm leaves the relationship with probability ψ_C (poaching threat) only when a lender denies a guarantee. If the relationship survives, the lender values it at ψ_F per unit of outstanding debt. Lenders differ in the cost of originating guaranteed loans, c_j^g per unit of guaranteed lending, with $c_M^g > c_B^g$, and along two further dimensions specified below.

Assumption 2 (Specialization) *The MFI sector serves a clientele with segment weights $(\rho, 1 - \rho)$ over (h, ℓ) , with $\rho \in (\frac{1}{2}, 1]$; the bank sector with weights $(1 - \rho, \rho)$. Within each opacity segment, the joint distribution of (n, b) is identical across lenders.*

Assumption 3 (Information) *Let $p_j(o)$ be the probability that lender j observes an opacity of client's revenue, independently of (n, b) . Opaque firms are read through relationship technology, so $p_M(h) > p_B(h) \geq 0$, while transparent revenues are hard information, so $p_j(\ell) = 1$ for every j .*

3.3 Private and Social Rankings

Private lenders rank their clients and fill their quota of loan guarantees from the top of this ranking. On the one hand, transparent and observed opaque clients are ranked by the index Ω_i (see Appendix A.1):

$$\Omega_i = T_i + P(g) s_{1,i} \quad \text{where} \quad P(g) \equiv \frac{B(g)}{\alpha - \beta}$$

with $\alpha \equiv 1 + \psi_F - \delta$, $\beta \equiv \psi_C \psi_F$, and $B(g) \equiv \beta + \varphi[r_g - \delta - g(1 - \delta)]$.⁷ On the other hand, unobserved opaque firms are pooled at their b -weighted mean $\bar{\Omega}_h$ and granted pro rata. Information thus moves an observed opaque client above the pool (a *promotion*) or below it (a *demotion*).

The social value per Sol of a guarantee to firm i through lender j is

$$SV_{ij} = \frac{T_i}{\varphi} - (1 - \delta)(1 - s_{1i}) - c_j^g - (\varsigma - 1)g(1 - \delta)(1 - s_{1i})$$

This is the debt saved, net of real default losses on the new loan, the origination cost, and the deadweight cost of the payout, where ς is the shadow cost of public funds (see Appendix A.2).

The planner ranks clients by:

$$S_i = T_i + K(g) s_{1i}, \quad \text{where} \quad K(g) \equiv \varphi(1 - \delta)[1 + (\varsigma - 1)g]$$

⁷We assume parameters such that $\alpha - \beta > 0$ and $B(g) > 0$ throughout.

The Coverage Threshold. The two rankings weight the same objects but differ in the survival probability weights $P(g)$ and $K(g)$, which move oppositely in the coverage rate. A higher g shields lenders from default losses (P falls) while making each default more costly in fiscal terms (K rises).

Proposition 1 (Coverage threshold) *If $P(0) > K(0)$ and $P(1) < K(1)$, there is a unique $\tilde{g} \in (0, 1)$ with $P(\tilde{g}) = K(\tilde{g})$ at which the private and social rankings coincide. For $g < \tilde{g}$ the private ranking overweights treated survival (inefficiently conservative); for $g > \tilde{g}$ it underweights it (inefficiently risky).*

The following results assume the risky-side inequality at the observed coverage rate, which the quantitative model quantifies rather than imposes.

Screening lets a lender act on comparisons that would otherwise be pooled, but each reallocation is evaluated with the private weight $P(g)$, not the social weight $K(g)$. Above $\tilde{g}$, a promotion that trades survival for treatment effect can be privately profitable yet destroy social value, granting a fragile, high-treatment client at the expense of a solid survivor.⁸ Since coverage in programs like Reactiva is close to one, the empirically relevant case is the risky side of the threshold: whether MFIs' superior information improves or reduces welfare is a quantitative question.

3.4 Welfare and MFI Participation

Aggregate welfare is $W = \sum_j \sum_{i \in \mathcal{G}_j} SV_{ij} \varphi b_i$, with $\mathcal{G}_j$ the set of firms granted by lender j . Let $S_j^m(Q_j)$ be the social index of lender j 's marginal guaranteed client at quota Q_j . Then, we can express the corresponding value per Sol $SV_j^m(Q_j) = S_j^m(Q_j)/\varphi - c_j^g - K(g)/\varphi$, and $S_j^{m,0}(Q_j)$ which is the same object in an *uninformed* scenario that shuts down the lender's screening,

⁸A promotion of a over b is privately profitable iff $\Delta T + P(g)\Delta s > 0$ and reduces welfare iff $\Delta T + K(g)\Delta s < 0$, with $\Delta T \equiv T_a - T_b$, $\Delta s \equiv s_{1,a} - s_{1,b}$.

making all opaque clients pooled into a single block at $\bar{\Omega}_h$ and granted pro rata. Since each lender fills from the top of its ladder, aggregate welfare is the integral of the marginal values, $W(Q_M) = \int_0^{Q_M} SV_M^m(q) dq + \int_0^{Q-Q_M} SV_B^m(q) dq$ (see Lemma A.1 in Appendix A).

Proposition 2 (Marginal decomposition) *Suppose the distribution of (n, b) is atomless. At every participation $Q_M = \gamma_M Q$ where the marginal schedules are continuous, the welfare effect of reallocating one Sol from banks to MFIs is*

$$W'(Q_M) = \underbrace{\mathcal{C}(Q_M)}_{\text{composition}} + \underbrace{\Delta_M(Q_M) - \Delta_B(Q - Q_M)}_{\text{precision}} - \underbrace{(c_M^g - c_B^g)}_{\text{cost gap}},$$

where $\mathcal{C}(Q_M) \equiv [S_M^{m,0}(Q_M) - S_B^{m,0}(Q - Q_M)]/\varphi$ and $\Delta_j(Q_j) \equiv [S_j^m(Q_j) - S_j^{m,0}(Q_j)]/\varphi$.

The composition margin $\mathcal{C}$ compares the two lenders' uninformed marginal guarantee granted and is driven by which segment each marginal guarantee falls in. For example, if the MFI marginal client is opaque and the bank marginal is transparent, the composition channel nets the higher treatment effect of opaque firms against their survival probability disadvantage.

The precision margin Δ_j measures how information changes the social index of the marginal grant. Its sign is not pinned down by the framework. Information promotes some clients and demotes others, and Δ_j nets these movements at the marginal client. At $g = \tilde{g}$ the private and social rankings coincide and information is unambiguously valuable (see Lemma A.2 in the Appendix). Away from the threshold, privately profitable promotions can reduce social value. Thus, whether the precision channel can be negative despite correct information is an empirical question that we address with our calibrated model in Section 6.

The participation result follows from capacity. The composition and precision advantages are attached to the MFI's opaque clientele, who are smaller borrowers, limiting the size of these channels. Appendix A.5 presents closed-form results on the sign of these channels under

one additional simplification, *segment ordering*, which assumes that opaque clients rank above transparent ones on the private ladder. Thus, once both lenders’ quotas exceed their opaque portfolios, the two marginal clients are drawn from the same transparent distribution and only the cost gap remains, which operates against MFIs. Welfare is therefore maximized at an interior $\gamma_M^* \in (0, 1)$. Segment ordering is a simplification that delivers closed forms, not a requirement for the mechanism. Our quantitative model, which imposes no ordering, exhibits and quantifies these forces.

The framework delivers the key objects we measure in the rest of the paper, namely treatment effects, which decline with firm-size and are larger among opaque firms, and the survival probability of guaranteed clients, which nets composition against selection. We next identify these channels in the data and feed them into a calibrated model, where we characterize the optimal MFI participation rate.

4 Empirical Framework

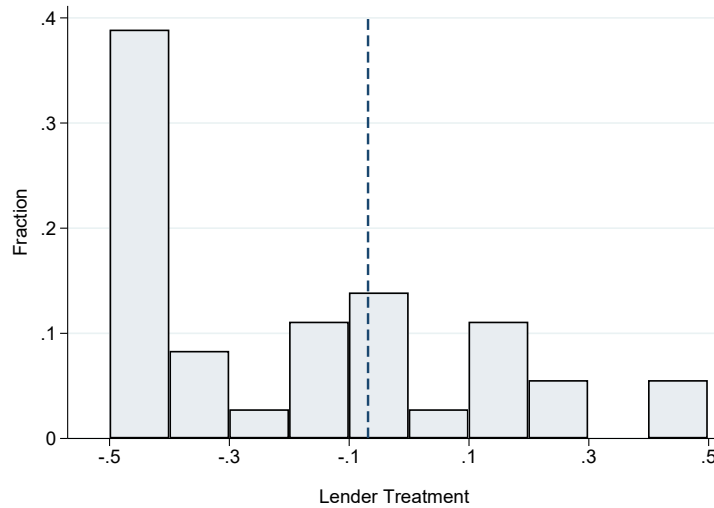
In this section, we describe our empirical approach to estimate the causal effects of loan guarantees and identify sensitive versus inframarginal borrowers. We exploit differences in lenders’ uptake of loan guarantees to estimate the effect of the program on credit supply. We compute the share of loan guarantees distributed by lender j , normalizing by lender j ’s pre-existing share of small business loans, as follows:

$$\text{Treatment}_j = \frac{\text{Share of Guaranteed Loans}_{j,2020} - \text{Share of Total Loans}_{j,2019}}{\text{Share of Guaranteed Loans}_{j,2020} + \text{Share of Total Loans}_{j,2019}} \times 0.5 \quad (1)$$

This continuum measure of treatment captures lender-specific reimbursement shocks (Granja et al. (2022)). Figure 2 plots the distribution of Treatment_j across lenders. We can see large heterogeneity in take-up. The dashed line indicates the median of bank treat-

ment, weighted by pre-recession market share.

Figure 2: Distribution of Bank Treatment



This figure plots the distribution of bank treatment measured by equation 1. The dashed line indicates the weighted-by-size median treatment.

Bank-firm level specification. We identify the credit supply effect of loan guarantees using a difference-in-differences approach that compares the outstanding debt that firms hold with more treated banks relative to less treated ones, before and after the program. We estimate the following equation:

$$Y_{ijt} = \theta \times \text{Treatment}_j \times \text{Post}_t + \delta_{ij} + \delta_{it} + u_{ijt} \quad (2)$$

Figure F2 splits financial institutions into high- and low-treatment groups based on median exposure and plots the evolution of credit and delinquency rates. Panel (a) shows that treatment was uncorrelated with aggregate credit growth before the program, while high treatment predicts a rapid and persistent expansion of credit thereafter.

Panel (b) shows that treatment is also uncorrelated with delinquency before the program. Finally, while highly treated MFIs exhibit a strong and persistent reduction in delinquency

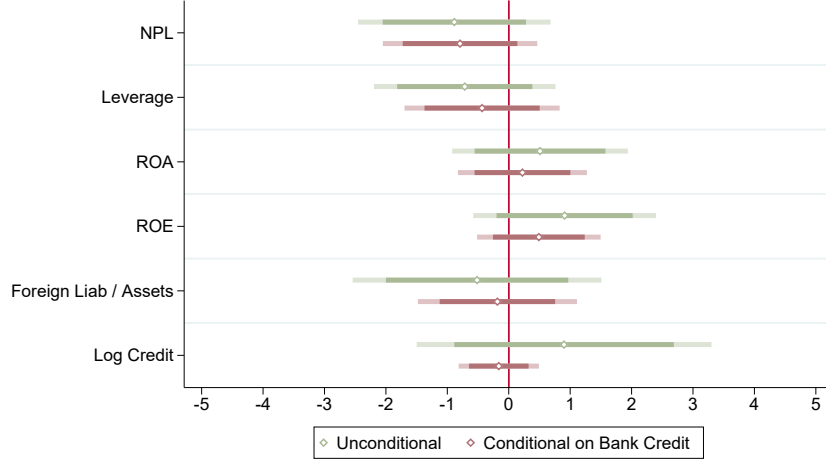
rates, traditional banks' delinquency remains constant during the recession and deteriorates after the grace period. The 3 percentage point difference in delinquency among highly treated MFIs and traditional banks after the recession is economically meaningful, as small business aggregate delinquency was around 7 percent before the recession.

where Y_{ijt} denotes the balance of total loans (in logs) that firm i has with lender j in period t , and Treatment_j is the standardized treatment defined by equation (1). We include firm-bank fixed effects δ_{ij} to control for match-specific time-invariant characteristics such as lender specialization in a given industry. δ_{it} denote firm-by-period fixed effects and remove any time-varying shock at the firm level. Our parameter of interest θ measures the impact of one standard-deviation higher treatment on bank lending to the same firm. Standard errors are clustered at the bank level.

Our identifying assumption is that absent the program, credit provided by more and less treated banks would have followed parallel trends. A remaining concern, even absent pre-trends, is that larger lenders may systematically serve larger firms better positioned to deal with COVID-19 restrictions. We address this by including time-varying fixed effects for bank and firm size-bins $\delta_{g(i),g(b),t}$, which allow us to compare credit obtained from more versus less treated banks within the same size group.

While our identification strategy does not require that more and less treated banks be similar in levels, such similarity makes the parallel trend assumption more plausible. We conduct a covariate balance test regressing multiple bank-level pre-determined characteristics, separately, on our measure of treatment. Figure 3 shows the estimated coefficients and corresponding confidence intervals indicating that treatment is statistically uncorrelated with the share of non-performing loans, leverage ratio, ROA and ROE, and foreign liabilities.

Figure 3: Balance of covariates



Note: This figure reports the results of a covariate balance test. It shows coefficient estimates along with 95% confidence intervals (darker bars) and 99% confidence intervals (lighter bars) for the impact of a one standard deviation increase in bank treatment across different variables. All variables are standardized to have zero mean and a standard deviation of one. The conditional regressions include fixed effects for each bank size quartile.

Firm level specification. We aggregate our dataset at the firm level to estimate the impact of the program on small firm performance. To do so, we construct the following measure of treatment:

$$\text{Treatment}_i = \sum_j \frac{L_{ij}}{L_i} \times \text{Treatment}_j \quad (3)$$

where L_{ij} denotes the outstanding debt that firm i holds with lender j in December 2019 and Treatment_j is defined in equation (1). Then we quantify the impact of the program on small firm performance by estimating the following equation:

$$Y_{it} = \beta \times \text{Treatment}_i \times \text{Post}_t + \delta_i + \delta_{x(i)t} + u_{it} \quad (4)$$

where Y_{it} denotes the balance of credit (in logs) or the delinquency rate of firm i in period t . We include firm fixed effects δ_i to absorb time-invariant heterogeneity. $\delta_{x(i)t}$ denotes time-

varying fixed effects for the vector $x(i)$ of firm characteristics including city, industry, risk category, age-bin, and size-bin based on pre-Covid debt, absorbing demand shocks at each of these levels. Standard errors are clustered at the main-lender level.

5 Results

We organize our empirical results in three blocks. First, we estimate the effects of the program on credit supply. Second, we analyze the response of firm performance. Finally, we conduct a heterogeneity analysis to identify the most sensitive firms and study how different lenders allocate guarantees.

5.1 Loan Guarantees and Credit Supply

We start by estimating the effect of the program on credit supply. We estimate equation (2) using the log of total loans as the dependent variable. Our results are reported in columns 1 to 3 of Table 3. We find that one standard deviation higher treatment leads to a 10% increase in credit supply in our benchmark specification reported in column 2. Our results are robust to using different sets of fixed effects as reported in columns 1 and 3.

Figure 4 plots event study graphs for the response of credit supply, showing quarterly treatment effects normalized to zero in the quarter before implementation. Treatment had null effects before the policy, consistent with our identifying assumption.

Moreover, our estimated coefficients are statistically indistinguishable from zero up to the first quarter of the policy, two quarters into the recession and before any meaningful volume of guarantees was distributed. Even though our identifying assumption is not testable by definition, this evidence provides critical support to our identification because it indicates that treatment was orthogonal to factors affecting firm credit demand when the shutdowns were already in place for two quarters. The balance of loans experience a significant and

persistent increase since the third quarter of 2020.

Table 3: Effect of Loan Guarantees on Bank Credit Supply

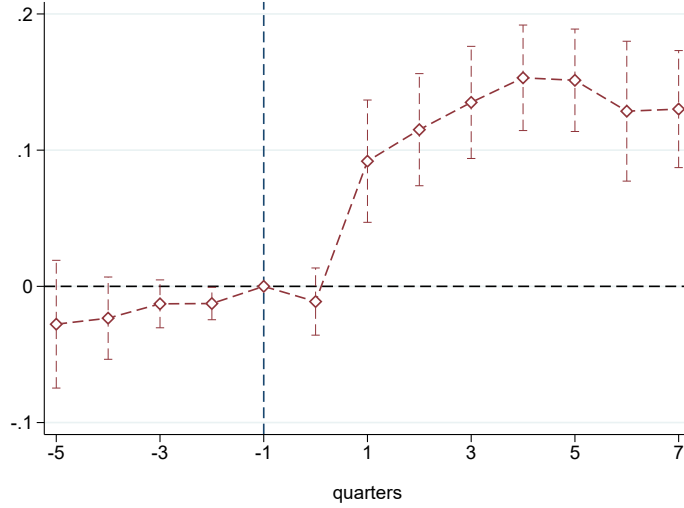
	<u>Total Loans</u>			<u>Non-Guaranteed Loans</u>		
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment _j × Post _t	0.163*** (0.050)	0.095** (0.047)	0.088*** (0.019)	-0.501*** (0.148)	-0.300*** (0.110)	-0.133*** (0.043)
Observations	37.4M	22.1M	22.1M	36.8M	21.5M	21.5M
Fixed Effects						
Bank-Firm	Yes	Yes	Yes	Yes	Yes	Yes
Quarter	Yes	–	–	Yes	–	–
Firm × Quarter	No	Yes	Yes	No	Yes	Yes
$g(i) \times g(b) \times \text{Quarter}$	No	No	Yes	No	No	Yes

This table shows the effect of the program on the balance of total loans and non-Covid-19 loans at the bank-firm level. Treatment is standardized. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. Standard errors are clustered at the bank level. Observations are expressed in millions.

We next examine whether government-backed guarantees crowd out normal bank activity (Stiglitz (1993), La Porta, Lopez-de-Silanes, and Shleifer (2002), Ru (2018)). We estimate equation (2) using the log of normal loans, those that are not collateralized by the government, as our dependent variable. We report our results in columns 4 to 6 of Table 3. We estimate that one standard deviation higher treatment leads to a decline of 30% in the supply of normal loans in our benchmark specification reported in column 5.

Figure F3 plots the event study graph associated with our benchmark specification showing no evidence of pre-trends. The balance of normal loans exhibit a steady decline after the program, indicating that loan guarantees partially crowded out the normal activity of banks. Thus, the program of guarantees fully explains the expansion of total credit during the recession.

Figure 4: Effect of the Program on Total Credit



This figure plots the quarterly effects of the program on total credit and non-Covid loans at the bank-firm level. The dependent variable is in logs. The program is implemented in the second quarter of 2020. Each dot is the coefficient on the interaction of treatment and quarter fixed effects. We normalize the treatment effect at the quarter right before the implementation of the program to be equal to zero. Confidence intervals at 95%.

5.2 Loan Guarantees and Small Firm Performance

To study the impact of loan guarantees on small firm performance, we aggregate our data at the firm level and calculate a corresponding measure of treatment as described in equation (3), which captures how well connected small firms were with more treated banks before the program. While the program led to an expansion of credit provided by highly treated banks, this does not mean better connected firms will receive more credit. If lending relationships were fully flexible, firms would easily switch to highly treated lenders and obtain guaranteed loans regardless of their pre-recession lender. Otherwise, better connected firms would receive more credit relative to less well connected ones, reflecting the stickiness of lending relationships.

We estimate equation (4) using total loans as our dependent variable. Table 4 (columns 1–

3) shows that firms with one standard deviation higher treatment experience a 14% increase in total loans after the program.⁹ We report quarterly treatment effects in panel (a) of Figure F4 showing no pre-trends. Similar to our bank-firm estimation, our treatment effects are statistically null up to the first quarter of the program, where COVID-19 shutdowns were in place for two periods but only a negligible part of guarantees was distributed. We find that better connected firms have more credit, and this effect is persistent. Our results indicate that lending relationships play a key role in shaping the ability of firms to obtain guaranteed loans.

Table 4: Effect of Loan Guarantees on Small Firm Borrowing

	<u>Total Loans</u>			<u>Non-Guaranteed Loans</u>		
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment _{<i>i</i>} × Post _{<i>t</i>}	0.163*** (0.038)	0.141*** (0.022)	0.140*** (0.024)	-0.433*** (0.074)	-0.436*** (0.076)	-0.357*** (0.080)
Observations	24.5M	24.5M	12.1M	24.2M	24.2M	12.0M
Fixed Effects						
Firm	Yes	Yes	Yes	Yes	Yes	Yes
$x^{\text{Real}}(f) \times \text{Quarter}$	Yes	Yes	Yes	Yes	Yes	Yes
$x^{\text{Fin.}}(f) \times \text{Quarter}$	No	Yes	Yes	Yes	No	Yes

This table shows the effect of the program on the balance of total loans and non-Covid-19 loans at the bank-firm level. Treatment is standardized. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. Standard errors are clustered at the bank level. Observations are expressed in millions.

We report the response of non-guaranteed loans in columns 4 to 6 of Table 4. Firms with one standard deviation higher treatment have a 36% lower balance of non-guaranteed loans relative to less well connected firms. Panel (b) of Figure F4 reports quarterly treatment effects, showing no evidence of pre-trends.

We now explore the response of delinquency rates, our measure of small firm performance.

⁹Since we cannot include firm-time fixed effects in this specification, the corresponding bank-firm level estimation is reported in column (1) of Table 3. Thus, a smaller coefficient in our firm-level specification means that firms have positive but limited ability to switch lenders.

We define delinquency as an indicator variable equal to one if the firm experiences more than 30 days of repayment delay in a given quarter. Our results are reported in columns 1 to 3 of Table 5. We find that highly treated firms perform better after the program. One standard deviation higher treatment reduces the probability of experiencing repayment delays by 2 ppts. Figure F5 plots the event-study graph associated with our benchmark specification and shows no evidence of pre-trends. Furthermore, the decline in delinquency among highly treated firms persists after the end of the grace period.

We then estimate two elasticities. First, we estimate the response of firm delinquency to program participation. Second, we measure the response of delinquency to credit received from the program. These elasticities identify which firms are sensitive versus inframarginal, and allow us to quantify how reallocating guarantees across firms and lenders affects the program's aggregate effectiveness. We estimate the following IV difference-in-differences specification:

$$\begin{aligned} \text{Delinquency}_{it} &= \gamma \times X_{it} + \delta_i + \delta_{x(i)t} + u_{it} \\ X_{it} &= \beta \times \text{Treatment}_i \times \text{Post}_t + \delta_i + \delta_{x(i)t} + u_{it} \end{aligned} \tag{5}$$

where X_{it} can be either being a program recipient or total loans. In each case, we instrument this variable with our firm-level measure of treatment. Our coefficient of interest γ measures the elasticity of delinquency to participating in the program or to the liquidity received from the program.

We report our results in columns 4 and 5 of Table 5. We find that program recipients are 11 percentage points less likely to exhibit repayment delays compared to non-participants. Furthermore, a 10 percent increase in credit reduces the probability of delinquency in 1.4 percentage points.

Table 5: Effect of Loan Guarantees on Small Firm Financial Performance

	OLS			IV	
	(1)	(2)	(3)	(4)	(5)
$\text{Treatment}_i \times \text{Post}_t$	-0.020*** (0.004)	-0.019*** (0.004)	-0.020*** (0.005)		
$\text{Recipient}_i \times \text{Post}_t$				-0.107*** (0.022)	
$\ln \text{Loans}_{it}$					-0.135*** (0.033)
Observations	24.6M	24.6M	12.1M	24.6M	24.6M
Fixed Effects					
Firm	Yes	Yes	Yes	Yes	Yes
$x^{\text{Real}}(f) \times \text{Quarter}$	Yes	Yes	Yes	Yes	Yes
$x^{\text{Fin.}}(f) \times \text{Quarter}$	No	Yes	Yes	Yes	Yes

This table shows the effects of being better connected to treated banks on the balance of total loans, non-Covid-19 loans, and delinquency rates at the firm level. Treatment is standardized. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. Standard errors are clustered at the firm level.

Overall, the program expanded credit supply and improved small firm performance. Highly treated firms received more credit and were less likely to be delinquent after the program. The lower delinquency is consistent with the unprecedented demand for external financing generated by COVID-19 restrictions, which outweighs potential firm risk-shifting incentives and reduced lender screening.

5.3 Heterogeneous Effects and Loan Guarantees' Allocation

In this subsection, we estimate the heterogeneous effects of the program and study the role of MFIs in allocating loan guarantees towards more sensitive firms. First, we split firms into two groups based on their outstanding debt in 2019. We define firms in the top decile of the debt distribution as large borrowers, and those in the bottom deciles as small borrowers.¹⁰

¹⁰It is worth noting that we split firms within the segment of small-firm credit. Thus, all firms in our sample are small.

Large firms account for 60 percent of small-firm debt in the pre-recession period. Then, we estimate equation (5) for each group of firms.

Table 6: Heterogeneous Effect of Loan Guarantees on Small Firm Financial Performance - Pre-Covid Debt Deciles

	Bottom Deciles (40% of Loans)			Top Decile (60% of Loans)		
	OLS	IV		OLS	IV	
	(1)	(2)	(3)	(4)	(5)	(6)
$\text{Exposure}_i \times \text{Post}_t$	-0.020*** (0.004)			-0.008* (0.004)		
$\text{Treated}_i \times \text{Post}_t$		-0.121*** (0.024)			-0.034* (0.019)	
$\ln \text{Loans}_{it}$			-0.151*** (0.035)			-0.039* (0.020)
Observations	21.7M	21.7M	21.7M	2.9M	2.9M	2.9M
Fixed Effects						
Firm	Yes	Yes	Yes	Yes	Yes	Yes
$x^{\text{Real}}(f) \times \text{Quarter}$	Yes	Yes	Yes	Yes	Yes	Yes
$x^{\text{Fin.}}(f) \times \text{Quarter}$	Yes	Yes	Yes	Yes	Yes	Yes

This table shows the effects of being better connected to treated banks on the balance of total loans, non-Covid-19 loans, and delinquency rates at the firm level. Treatment is standardized. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. Standard errors are clustered at the firm level.

Our key hypothesis is that liquidity needs are bigger among small firms. Since large firms typically exhibit higher levels of cash-flows and liquidity buffers, they will be more likely to survive the recession even without the program.¹¹ Our estimation results are reported in columns (1) to (6) of Table 6. Consistent with our hypothesis, the elasticity of delinquency to credit among smaller firms is four times that of large borrowers. While a 10 percent increase in credit reduces the probability of delinquency by 1.5 percentage points among smaller borrowers, the response is only 0.4 percentage points among large borrowers.

¹¹The alternative hypothesis discussed in our conceptual framework is that, in a deeper recession, large firms can be more sensitive to the program. This is because, given the size of the shock, small firms with low levels of cash will not survive even if they receive liquidity from the program.

Table 7: Heterogeneous Effect of Loan Guarantees on Small Firm Financial Performance - Industry Contact

	High-Contact			Low and Non-Contact		
	OLS	IV		OLS	IV	
	(1)	(2)	(3)	(4)	(5)	(6)
$\text{Exposure}_i \times \text{Post}_t$	-0.021*** (0.003)			-0.016*** (0.005)		
$\text{Treated}_i \times \text{Post}_t$		-0.116*** (0.020)			-0.088*** (0.025)	
$\ln \text{Loans}_{it}$			-0.162*** (0.035)			-0.093*** (0.028)
Observations	16.5M	16.5M	16.5M	8.0M	8.0M	8.0M
Fixed Effects						
Firm	Yes	Yes	Yes	Yes	Yes	Yes
$x^{\text{Real}}(f) \times \text{Quarter}$	Yes	Yes	Yes	Yes	Yes	Yes
$x^{\text{Fin.}}(f) \times \text{Quarter}$	Yes	Yes	Yes	Yes	Yes	Yes

This table shows the effects of being better connected to treated banks on the balance of total loans, non-Covid-19 loans, and delinquency rates at the firm level. Treatment is standardized. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively. Standard errors are clustered at the firm level.

Second, we split firms based on industry. Following Cerda et al. 2023, we define high-contact industries as those with customer-facing interactions (e.g., retail, restaurants, transportation), and low-contact industries as those where such interactions are limited (e.g., manufacturing). Our hypothesis is that high-contact industries faced a larger liquidity shock, as they were more directly affected by COVID-19 restrictions.

Our estimation results are reported in columns (1) to (6) of Table 7. Consistent with our hypothesis, the elasticity of delinquency to credit is bigger in high-contact industries. While a 10 percent increase in liquidity reduces the probability of delinquency by 1.6 percentage points in high-contact industries, the effect is 0.9 percentage points in low-contact sectors.

Having identified the most sensitive borrowers, those that are smaller and in high-contact sectors, we now study the allocation of guarantees implemented by each type of lender. The

top panel of Table 8 reports the participation of small and large borrowers in the portfolios of loan guarantees and pre-recession small business loans for traditional banks and micro-finance institutions.

Table 8: Share of Credit and Loan Guarantees by Type of Bank and Firm

	Loan Guarantees		Pre-Recession Loans	
	Banks (1)	MFIs (2)	Banks (3)	MFIs (4)
Size (pre-Covid Debt):				
Small	0.29	0.65	0.26	0.50
Large	0.71	0.35	0.74	0.50
Industries:				
High-Contact	0.46	0.60	0.59	0.64
Less- and Non-Contact	0.54	0.40	0.41	0.36
Lender share:				
	0.75	0.25	0.48	0.52

This table reports the composition of MFI and traditional bank portfolios. Columns (1) and (2) show the distribution of loan guarantees across firm types, while columns (3) and (4) show the distribution of pre-recession loans as of December 2019. Small firms are defined as those in the bottom nine deciles of the pre-Covid debt distribution among small businesses; large firms are in the top decile. High-contact industries include corner stores, restaurants and bars, hotels, bus passenger, clothing stores, and others as defined in Cerda et al. 2023. All figures represent shares within each lender type.

Traditional banks' portfolios are more concentrated among large borrowers, who represent 74 percent of bank small-firm credit before the recession, compared to 50 percent for MFIs. This specialization carries over to guarantee allocation: 71 percent of traditional bank guarantees went to large, low-sensitivity borrowers, while MFIs directed 65 percent of their guarantees toward smaller, highly sensitive borrowers.

Similarly, we observe that traditional banks provided more guarantees to low-contact industries, while MFIs attended mainly high-contact sectors. These allocation patterns

indicate that MFIs were more likely to provide guarantees to high-sensitive borrowers such as smaller firms in high-contact industries, mirroring their pre-recession specialization.

We conduct a back-of-the-envelope calculation, explained in more detail in Appendix C, to measure the gains from observed MFI participation using our reduced-form evidence. Given our estimates and the observed participation of MFIs and traditional lenders, the program reduced delinquency by 12 percentage points. In contrast, if all guarantees had been allocated through traditional banks, aggregate delinquency would have declined by only 8 percentage points.

6 Quantitative Model

We now build a quantitative model of the Peruvian credit market to quantify the forces described in Section 3. The model includes heterogeneous firms who choose whether to operate in the formal economy or not, and heterogeneous lenders that compete in prices under differentiated product demand and acquire costly information to serve informal firms.

6.1 Firms

A unit mass of risk-neutral firms is heterogeneous along two dimensions. Each firm draws a permanent productivity $\hat{z} \geq 1$ from a Pareto distribution with tail parameter ξ . Then, firms draw an opacity type σ . The pair $(\hat{z}, \sigma)$ is publicly observable and defines the firm's *class*. Opacity governs the variance of a transitory revenue shock u , which follows a log-normal distribution with $E[u] = 1$ and is realized after the loan terms are set; thus, it determines the probability of default in normal times.

Technology. A firm that borrows b units of capital at gross rate r operates a technology with segment-specific capital share α_s with $s \in \{I, F\}$ denoting informal and formal

segments, respectively. The formal technology is more capital-intensive ($\alpha_F > \alpha_I$). Firms produce $u \hat{z} b^{\alpha_s}$ and realized net cash after repayment is:

$$n(r, b, u; \alpha_s) = u \hat{z} b^{\alpha_s} - rb \quad (6)$$

Capital is committed before the transitory shock is known. The firm chooses a single loan size b per class and formality status to maximize its ex-ante expected utility, anticipating the menu of rates it will face.

Credit Demand. Each firm draws an i.i.d. Gumbel taste shock ε_{ij} over the $N_M + N_B$ individual lenders and selects the lender offering the highest value. Lender j posts a rate r_j ; the firm, knowing its transitory shock u , evaluates the net cash $n_{ij} = n(r_j, b, u)$ it would obtain under that rate, and values the lender at $U_{ij} = n_{ij}/\omega_s(\hat{z}) + A_s + \varepsilon_{ij}$, where $\omega_s(\hat{z})$ is a taste dispersion parameter that governs the elasticity of credit demand and A_s is a segment-specific amenity which captures the non-pecuniary benefits of formalization.¹² The firm's ex-ante objective is to maximize:

$$\mathbb{E}_{u, \varepsilon} \left[\max_j \left(n(r_j, b, u) / \omega_s + A_s + \varepsilon_{ij} \right) \right]$$

The optimal b therefore satisfies a fixed-point condition solved jointly with the credit-market equilibrium as we describe in more detail in Appendix B. All lenders of a type post the same rate in equilibrium, so the type-level MFI share in a cell is:

$$S_M = \frac{N_M e^{v_M}}{N_M e^{v_M} + N_B e^{v_B}}$$

¹²We define the taste dispersion as $\omega_s(\hat{z}) = \bar{\omega}_s (\alpha_s \hat{z})^{1/(1-\alpha_s)}$, i.e., we scale it by the frictionless optimal firm size so the price elasticity of credit demand is approximately constant across firms with different productivity. We also define $\bar{\omega}_s$ as segment-specific to capture remaining interest rate differences across segments. Note that A_F is jointly calibrated with c_e to match both the share of formal firms and formal borrowers.

with $v_j = n_j/\omega + A_s$, and $S_B = 1 - S_M$. An individual lender serves the share $\tilde{s}_j = S_j/N_j$.

Normal-times delinquency. After the loan is made, u realizes, and net cash is determined, the firm faces a residual liquidity shock ν_0 and survives normal times if $n > \nu_0$, where ν_0 has distribution Φ_0 , so the survival probability is $\Phi_0(n) = \min\{(n/c_0)^{\eta_0}, 1\}$ when $n > 0$, and zero otherwise. By governing the dispersion of the unobservable revenue shifter, opacity shapes the probability of default in normal times.

Relation to the conceptual framework. The firm block of the quantitative model relaxes two simplifications of the conceptual framework. First, opacity acts through the variance of the unobservable revenue component rather than a first-order stochastic dominance shift. Thus, whether opaque firms are more sensitive to guarantees is an equilibrium outcome, not an imposed restriction. Second, the pre-recession equilibrium rate r is an equilibrium object instead of being normalized to one, so the private value of a guarantee also reflects the actual loan margin.

6.2 Lenders

There are two lender types: N_M identical microfinance institutions (MFIs) and N_B identical traditional banks. Lender $j \in \{M, B\}$ has a constant per-unit lending cost c_j , with $c_M > c_B$. Since the firm commits to a loan size b before the transitory shock is known, the lender takes b as given when setting r . The expected per-unit margin on a u -type firm priced at r is:

$$\mu_j(r, b, u) = \Phi_0(n(r, b, u)) (r - \delta) + \delta - (1 + c_j) \quad (7)$$

where δ is the recovery value per unit lent and $n(r, b, u)$ is the net cash from (6).

Information and screening. The firm's class $(\hat{z}, \sigma)$ is publicly observable. The transitory shock u is the object of asymmetric information. Lenders have a segment-specific screening technology. By paying a per-applicant cost κ_{sj} , lender type j observes each applicant's u in segment s and can post a type-contingent rate schedule $r_{sj}(u)$. A lender that does not refine posts a single pooled rate for the class $(\tilde{z}, \sigma)$ and serves the resulting set of u -types. MFIs have a screening advantage, $\kappa_{sM} < \kappa_{sB}$.

The formal segment is thus the same screening-and-pricing game as the informal segment, but run with a different parameter configuration that captures the informational benefit of being formal. In our calibration below, we set screening costs to zero in the formal segment, so asymmetric information is not a binding force.

Pricing. In each segment, competition proceeds in two layers. First, lenders choose whether to refine or not. Second, lenders set rates to maximize per-borrower profits against the rival's offer. A refined lender solves a first-order condition for each u -type, accepting only those with non-negative margin; a pooled lender solves an expected first-order condition and participates only if its expected class profit is non-negative.

The resulting payoffs define a screening game. At the calibrated parameters this game has a unique pure-strategy Nash equilibrium in the overwhelming majority of classes (99.7%). Where it does not, the equilibrium that maximizes total surplus is selected (see Appendix D for diagnostics and a sensitivity analysis).

6.3 Formality

Formality is chosen at entry, when the firm knows only its potential productivity $\hat{z}$. If the firm chooses the informal segment, it draws an opacity type $\sigma \in \{\sigma_L, \sigma_H\}$ with probabilities $(m_L, 1 - m_L)$. If it chooses the formal segment, its opacity is σ_F . The pair $(\hat{z}, \sigma)$ then becomes publicly observable for the credit market. The firm compares the ex-ante expected value of

operating in each segment, taking as given the credit-market equilibrium and the segment-specific technology. The firm becomes formal if the expected gain covers the formalization cost c_e :

$$\Delta(\hat{z}) \equiv V_F(\hat{z}) - V_I(\hat{z}) \geq c_e \quad (8)$$

where $V_F(\hat{z})$ and $V_I(\hat{z})$ are the ex-ante expected inclusive values that the firm would obtain in the formal and informal segments, respectively. Two forces make the value gap increasing in $\hat{z}$. First, the formal technology is more capital-intensive ($\alpha_F > \alpha_I$), so the surplus from operating at formal rates grows faster with productivity than the informal surplus. Second, the lower opacity σ_F reduces the residual risk faced by formal firms, further raising their expected value relative to the informal segment.

6.4 Equilibrium and Calibration

Equilibrium. The credit-market equilibrium consists of three pieces. For every class $(\hat{z}, \sigma)$, a committed loan size b , a screening profile $(a_M, a_B) \in \{0, 1\} \times \{0, 1\}$, and a rate schedule $r_j(u)$ (if refined) or a single pooled rate r_j (if not refined) for each lender type, together with a service rule that determines which u -types are served. A formality status completes the equilibrium.

These objects satisfy five conditions. First, in each segment the committed size b maximizes the firm's expected value. Second, in every screening regime of either segment, active lenders set rates to maximize profit given the rival's offer (pointwise for each u -type when refined, and as a single pooled rate otherwise) with the service rule that a type is served if and only if the margin is non-negative (refined), and a class is served if and only if expected class profit is non-negative (pooled). Third, the screening regime in each class is a pure-strategy Nash equilibrium of the screening game (selected by total surplus when multiple or no pure-strategy equilibrium exists). Fourth, within every served cell borrower alloca-

tion across lender types follows the logit share evaluated at equilibrium net cash. Fifth, the formality assignment satisfies the cutoff rule (8).

The committed loan size and the credit-market equilibrium are solved jointly class-by-class and segment-by-segment. Then, the formality cutoff is determined among the resulting menus.

Calibration. The model is calibrated to match a set of economy-wide moments from the Peruvian small-firm credit market and the Reactiva program. Pre-recession parameters (i.e., lending and screening costs, the formalization cost, the Pareto tail, and the normal-time liquidity shock distribution) are calibrated internally. The targeted moments include the informal borrower share (around 80%, proxied by the share of registered borrowers that are natural persons operating businesses), the MFI share of total borrowers by number and value, average interest rates of informal and formal borrowers (net rates of approximately 60% and 25%, respectively), average interest rates of each type of lender (50% and 30% for MFIs and banks, respectively), and the pre-COVID delinquency rate (7% range). The full parameter set and the fit of every moment are reported in Appendix C.

6.5 The Guarantee Program

After the credit market clears, an unexpected aggregate recession hits. Recession survival is governed by a second survival function $\Phi_R(n) = \min\{(n/c_R)^{\eta_R}, 1\}$ for $n > 0$, and zero otherwise. A guaranteed credit,¹³ equal to a fraction φ of the firm's loan, is distributed through private lenders so firms' recession survival moves from $s_0 = \Phi_R(n)$ to $s_1 = \Phi_R(n + \varphi b)$, and the treatment effect of the guarantee on the firm is $T = s_1 - s_0$.

The government provides a total guarantee volume Q , allocating a share γ_M of this volume

¹³Parameters such as the guaranteed rate or the size of the program are assigned directly from institutional features of Reactiva, while the recession shock is calibrated to reproduce the estimated treatment-effect heterogeneity. See Appendix C for details.

to MFIs and the remainder to banks. Each lender type distributes guarantees exclusively among its own clients, ranking them by the private value Ω_i which is given by:

$$\Omega_i = T_i \times (r_i + \psi_F - \delta + m_j) + s_{0,i} \times (\psi_C \psi_F + m_j)$$

where r_i is the equilibrium loan rate from the pre-recession credit market, T_i and $s_{0,i}$ are the firm's treatment effect and baseline survival probability, respectively, and $m_j = \varphi[r_g - \delta - g(1 - \delta)]$ captures the loan-guarantee-margin where r_g is the guaranteed-loan rate. Remaining parameters are the same as in the conceptual framework described in Section 3.

When a lender knows the individual client u (formal clients and refined informal) it uses the client's own Ω_i ; otherwise it uses the class-average $\bar{\Omega}$. Then, guarantees are allocated accordingly until the quota is filled.

Lender heterogeneity implies that the lender mix defined by γ_M has aggregate consequences that operate through two channels. First, the *composition* channel, which reflects that MFIs and banks serve different clienteles. MFIs specialize in the opaque, small firms whose survival probability responds most to receiving a guarantee. Second, the *precision* channel that reflects MFIs' superior information about informal, opaque borrowers, which lets them target guarantees within a class rather than by the class average.

Our main results isolate these channels by shutting down one at a time while holding the rest of the equilibrium fixed. We let the *composition* channel always active, while the *precision* channel is removed by forcing all informal lenders to allocate guarantees using class-average private values Ω regardless of their screening decisions. This eliminates the advantage of refined individual knowledge while keeping the same clientele pools. The cumulative contribution of each channel provides a transparent decomposition of the gains from MFI participation.

6.6 Main Results

Debt saved from default. Figure 5 reports the total debt the program saves from default, as a percentage of aggregate outstanding debt, plotted against the MFI share γ_M . For each firm i , receiving a loan guarantee increases its recession survival probability by T_i , so an additional $T_i b_i$ *soles* of debt are repaid in expectation. Summing over guaranteed firms and dividing by the pre-recession stock of debt we have the debt saved from default, in percentage terms, relative to the non-program scenario. We decompose this gain into the composition and precision channels.

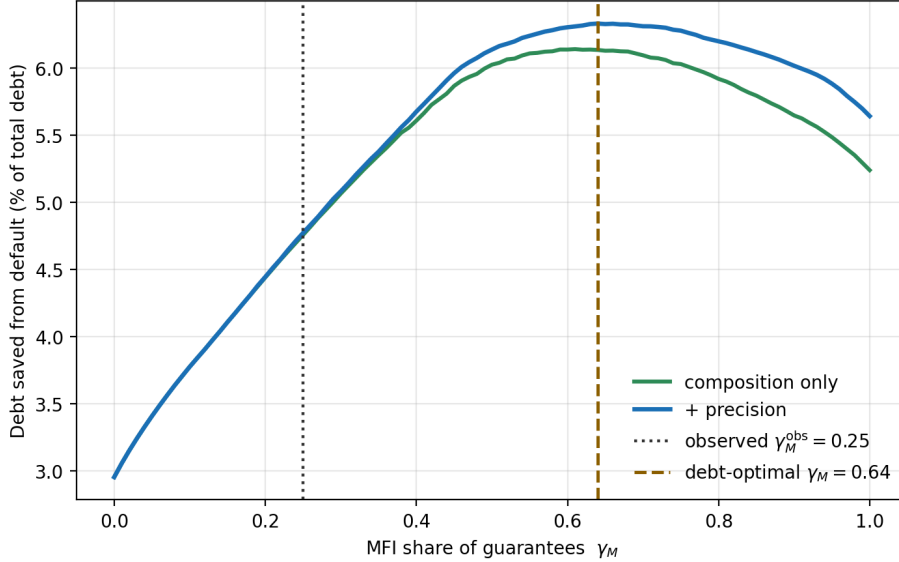


Figure 5: Debt saved from default as a percentage of aggregate outstanding debt, shown against the MFI share γ_M . The green curve uses composition only; the blue curve adds the precision channel. The dashed line marks the observed MFI share $\gamma_M^{\text{obs}} = 0.25$. Calibrated parameters are reported in Appendix C.

Under the composition channel alone (green curve), reallocating guarantees from banks to MFIs raises debt saved steeply at first, because MFI quotas are directed at opaque, high- T firms. The marginal gain diminishes as the MFI quota is distributed toward safer clients. The curve peaks at $\gamma_M = 0.6$, where the program saves 6.1% of debt, 2.4 percentage points

more than in a bank-only scenario.

Figure F7 plots the treatment effect by firm size among all eligible small borrowers. Smaller firms respond far more strongly to the guarantee. Those in the bottom 90% of the loan-size distribution have an average treatment effect of 0.11, while the largest 10% have 0.04. Guarantees bind where firms are small enough that the additional liquidity moves them across the survival margin. Figure F8 plots the average treatment effect by lender type, considering only firms that obtained a guaranteed loan at the observed MFI participation $\gamma_M^{\text{obs}} = 0.25$. MFI clients have a substantially higher average treatment effect (0.20) relative to bank clients (0.11). This difference in treated clienteles explains why reallocating guarantees toward MFIs raises debt saved from default. However, this gain is bounded once the MFI quota is large enough to have absorbed its high-treatment pool.

Adding the precision channel (blue curve) increases the maximum debt saved to 6.3%, and moves the optimal MFI participation to $\gamma_M = 0.64$. At the observed share $\gamma_M^{\text{obs}} = 0.25$, by contrast, the precision effect is negligible, since composition plays a predominant role at the beginning. Figure F9 unpacks the precision channel at the client level. For each refined MFI client we compare its own treatment effect T_i to the pooled average $\bar{T}_{\text{class}}$ of its class, which is exactly what the lender would face without screening. Then, we classify refined clients by the sign of the deviation. Screening is informative and efficiently directed. 59% of refined clients have above-average treatment effects, against 26% below. Moreover, the above-average group exceeds its class by 28% on average, while the below-average group falls short by 17%.

Our results indicate that, while a banks-only program ($\gamma_M = 0$) saves 3.7% of aggregate debt from default on its own, introducing MFIs at the observed share $\gamma_M^{\text{obs}} = 0.25$ raises this to 4.8% (i.e., a 29% increase in the value of debt saved from default relative to the banks-only implementation). Moving further to the debt-optimal share $\gamma_M = 0.64$ raises it to 6.3%, a further 33% increase over the observed allocation.

Social welfare. The social value of a unit of guaranteed volume allocated by lender j , given the equilibrium set of recipients, is:

$$SV_j = \frac{\bar{T}_j}{\varphi} - c_j^g - (1 - \delta)(1 - \bar{s}_{1,j}) - (\varsigma - 1)(1 - \delta)g(1 - \bar{s}_{1,j})$$

where $\bar{T}_j$ and $\bar{s}_{1,j}$ are allocation-weighted averages, c_j^g is the lender's guarantee-origination cost, and $\varsigma > 1$ is the social cost of public funds. The parameter g is the share of a defaulted guaranteed loan repaid by the government. Aggregate welfare at a given split γ_M is $W(\gamma_M) = \gamma_M Q \cdot SV_M + (1 - \gamma_M) Q \cdot SV_B$.

While the debt metric considers only the outstanding debt that is saved from default by the guarantee, SV_j subtracts the lender's guarantee-origination cost, the expected capital loss on the new guaranteed loan, and the dead-weight cost of financing the government.¹⁴

Figure F10 plots the social-welfare gain from MFI participation defined as $G(\gamma_M) = 100 [W(\gamma_M) - W(0)]/Q$. We compute it under the composition channel alone, composition plus precision, and the full set of channels that include the origination-cost gap.¹⁵

Under the composition channel alone, the welfare gain peaks at 5.7% of program volume at a participation $\gamma_M = 0.65$. Adding the precision channel leaves the welfare peak at 5.7% but reduces optimal participation to $\gamma_M = 0.62$. Finally, incorporating the origination-cost gap only reduces welfare gain to 5.1%.

Our calibrated economy generates a welfare-optimal MFI share close to the share at which debt saved is maximized. Two forces shape the welfare optimum relative to the debt one. First, the survival composition of guaranteed clients raises it. MFI recipients survive the recession at higher rates than bank recipients ($\bar{s}_{1,M} = 0.87$ versus $\bar{s}_{1,B} = 0.84$, consistent with empirical averages). Second, the origination-cost gap partially offsets it. MFIs distribute

¹⁴The cost-gap is calibrated using the guarantee lending rate charged by each type of lender.

¹⁵The origination-cost channel is removed by setting the guarantee-origination cost of MFIs equal to that of banks. Since origination costs do not affect the private ranking they leave the allocation unchanged and only matter for welfare calculation.

guarantees at a higher unit cost than banks ($c_M^g = 0.035$ versus $c_B^g = 0.025$), a penalty that is absent from debt saved and that pushes the welfare optimum marginally below the debt optimum.

7 Conclusions

The global expansion of microfinance institutions in emerging markets has raised important questions about their capacity to support long-term development and facilitate short-term recovery during crises. Using a large-scale loan guarantee program in Peru as our empirical setting and a quantitative framework for counterfactual analysis, we provide evidence on how MFIs shape the effectiveness of financial stimulus in developing countries.

We show that loan guarantees increased credit supply and improved firm performance, with stronger effects among smaller firms and those in high-contact industries, the segments facing the most severe liquidity needs. MFIs played a central role in channeling guarantees to these high-sensitivity segments, reflecting their pre-recession specialization in relationship-based lending.

Our quantitative model shows that distributing guarantees solely through traditional banks would have saved only 3.7 percent of aggregate outstanding debt from default, whereas the actual allocation with MFI participation raised this to 4.8 percent. These gains reflect three forces: MFIs' specialization in the small, opaque borrowers most responsive to guarantees, the precision with which their screening directs guarantees within that clientele (whose value depends on the guaranteed rate), and their higher cost of originating guarantees. Our findings indicate that the mix of heterogeneous lenders plays a key role in shaping the effectiveness of financial stimulus in emerging markets.

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